



Introduction to Bioinformatics Project 1 - Read Mapping and Genome Assembly

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Part A - Downloading E. Coli WGS data, Preliminary Analyses, and Quality Controls:

Answer the following questions about the short reads fastq file (you can use existing R packages such as ShortRead, or similar Python packages):

- How many reads are in the fastq file?

2273185

- Print the identifier, quality, and sequence of the first read of the fastq file.

```
@SRR8185317.1 ERR022075.10741970 length=100
```

```
AGCGGTACACATTATGGGTCTGCTCTCCGCAGGCGGCGGTACACAGCCACGAAGATCACATCA  
TGGCGATGGTAGAACTGGCAGCTGAACGCGGCGCAGAA
```

```
+SRR8185317.1 ERR022075.10741970 length=100
```

```
IIGIIIIIIIIIIIIIIIIIIIIII@IHHEHIIIIIIIIIIIIIIIIIIIGIHHFIIIEIIGIHHGH@BE3BB>@>>2?@  
8?>?A@1
```

- How many times does the TTAAATGGAA subsequence appear in the file?

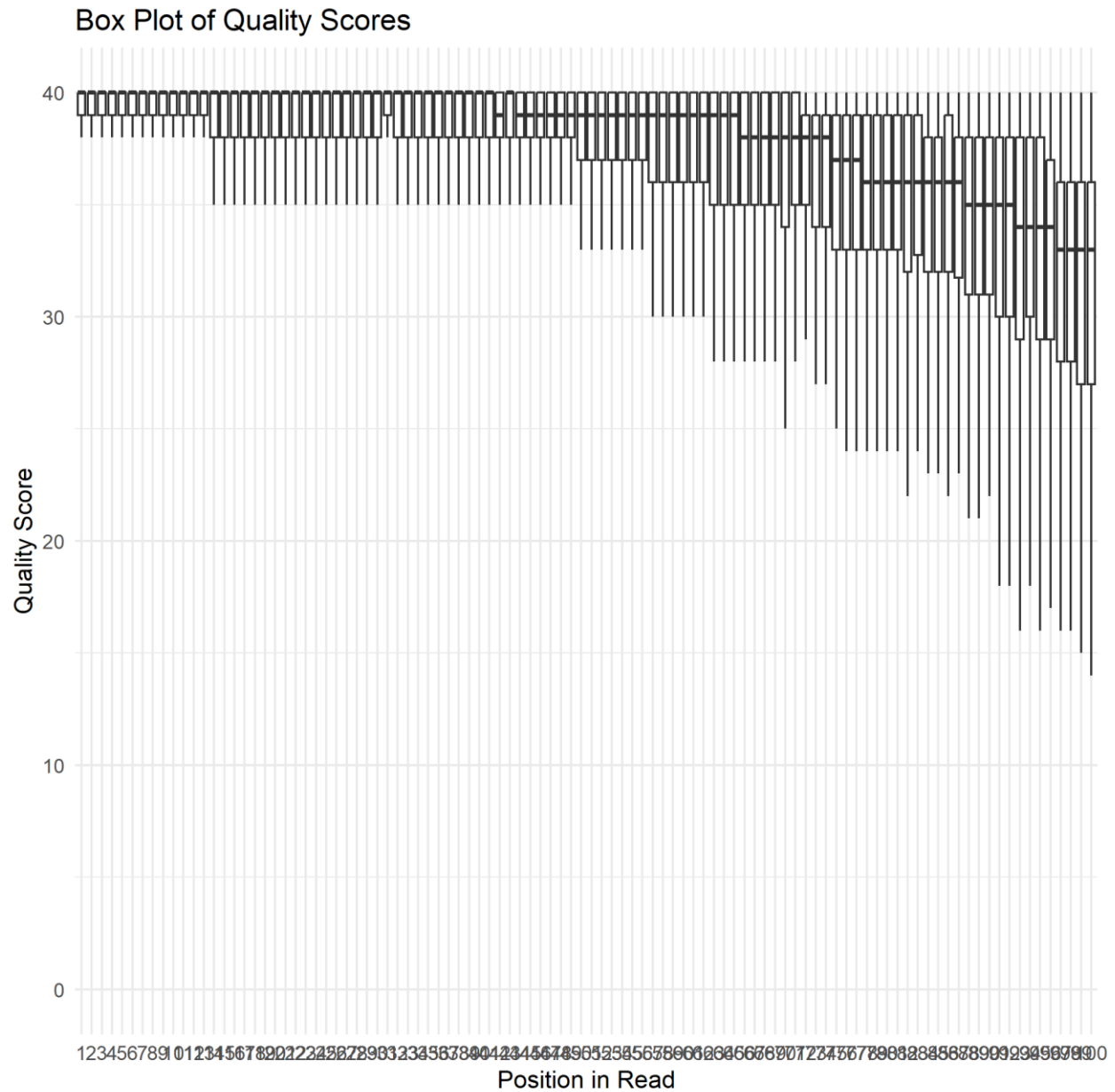
176

- Extract the first 1000 sequences of the fastq files (4000 lines).

It's in my qc_results folder.

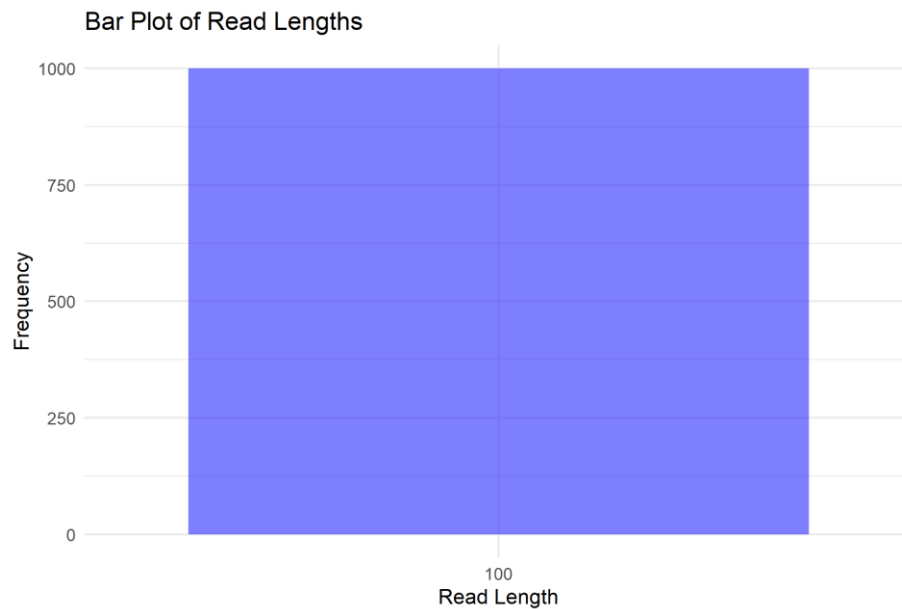
- Plot the quality of the reads in the fastq file using a box plot:

Note: I chose to use a subset of the original data for analysis primarily due to performance constraints. The original dataset was large, and when attempting to run the analysis, I encountered errors and program stoppages due to the excessive size of the data. By using a subset, I was able to manage the analysis more effectively, ensuring smoother execution while still obtaining relevant results.

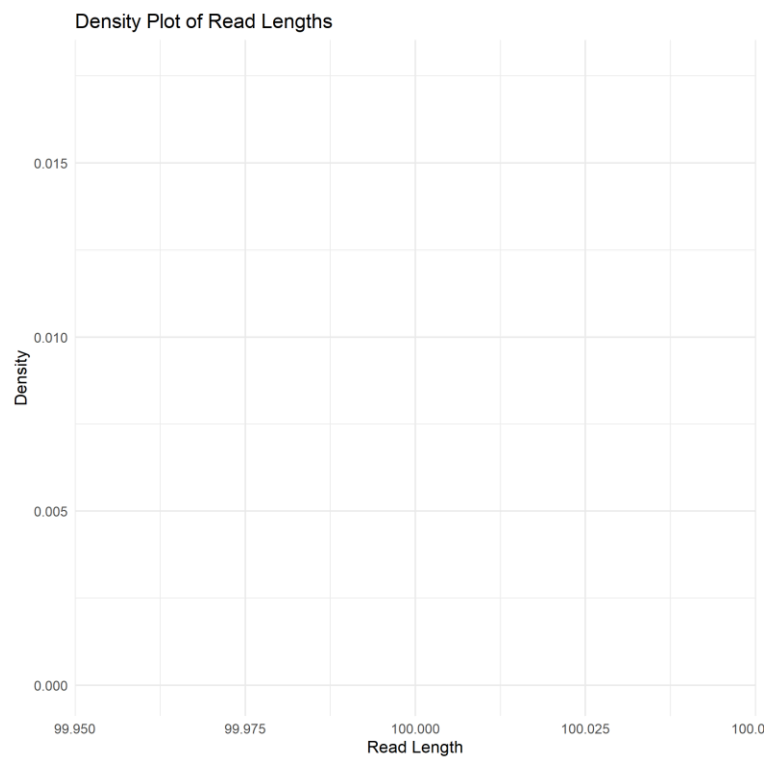


The box plot of quality scores reveals that most base positions in the sequencing reads have high-quality scores, with an interquartile range around 30-40, suggesting good sequencing quality. However, there is a noticeable decrease in quality towards the 3' end of the reads, which is consistent with the typical drop in quality often observed in Illumina sequencing.

- Show the distribution of read lengths using a density plot.



The bar plot of read lengths shows that the majority of the reads fall within the 100-200 base length category, which is typical for the sequencing technology used. This indicates that the sequencing run produced mostly uniform read lengths, with minimal presence of either very short or very long reads.



The density plot of read lengths indicates that the majority of reads are around 150-200 bases in length, with a relatively uniform distribution. This suggests that the sequencing run was of high quality, and most reads were of consistent length.

- Perform quality control of the reads using FastQC and interpret the results.

Note: Due to the large size of the original data, processing them directly caused errors and program interruptions. To address this issue, I created a subset of the data for analysis. Additionally, FastQC was unable to process the original dataset effectively due to their size. This subset approach ensured smooth processing and accurate quality control.

Part C - Read Mapping:

- Print the head of the obtained SAM file from the previous question. Explain what you see for the first hit (you can do this step either in Linux or R).

Mapping raw sequencing reads to a reference genome:

```
SRR8185317.1 0 NC_000913.3 3785592 60 100M * 0 0
AGCGGTACACATTATGGGTCTGCTCTCCGCAGGCGGCGTACACAGCCACGAAGATCACATCA
TGGCGATGGTAGAACTGGCAGCTGAACGCGGCGCAGAA
IIGIIIIIIIIIIIIIIIIIIIIII@IHHEHIIIIIIIIIIIIIIIIIIIGIHHFIIIEIIGIHHGH@BE3BB>@>>2?@
8?>?A@1 NM:i:0 MD:Z:100 AS:i:100 XS:i:0
```

Explanation:

SRR8185317.1: Read ID.

0: Mapping flag, indicating the read is aligned in the forward direction.

NC_000913.3: The reference sequence (chromosome) where the read aligns.

3785592: The position on the reference genome where the read aligns.

60: Mapping quality, indicating high confidence in this alignment.

100M: Read length (100 base pairs) aligned perfectly to the reference.

**AGCGGTACACATTATGGGTCTGCTCTCCGCAGGCGGCGTACACAGCCACGAAGATCACATCA
TGGCGATGGTAGAACTGGCAGCTGAACGCGGCGCAGAA:** Sequence of the read.

**IIGIIIIIIIIIIIIIIIIIIIIII@IHHEHIIIIIIIIIIIIIIIIIIIGIHHFIIIEIIGIHHGH@BE3BB>@>>2?@
8?>?A@1:** Quality scores for each base in the read.

NM:i:0: No mismatches in the alignment.

MD:Z:100: Entire read aligned perfectly (100% match).

AS:i:100: Alignment score (perfect alignment with score 100).

XS:i:0: No alternative alignment.

Alignment Results of the Assembled Genome with the Reference Genome:

```
NODE_6_length_153168_cov_18.165897 16 NC_000913.3 2572950 60 153168M * 0 0
CACTGGCAAGCGCGATATCCGCCAGCTGCGCCGCCGTCGGCTGTGGCACCACGCTGCAATCG
```

GCAAAGCCCAACGCCGGGCCGCTGTACTGTGGCAACATCAGGAAAATGGAAGAGAGCGTTT
TACAGCCTGGCTGCAAGCCGATAATGCGTAATCCGGCACGCAGCACATTTGCCGTGGAAGA
GAGGTTGCCCCGCGATACAGACATCCGCTTTACCGGCGCTGACCATTGCTGCGGCGAACATCA
GAGGGTCGGTAAGTTTTTCCAGCGCATCCGGCGGCGTTTTTTCGCCCCGCGCGGGCCAGCCAG
CGATGAGCAAATTCTT...GGTTATTGAGGAGCTTAAGGATAAATTTCTGGTAAGGAGGACACG
TATGGAAGTGGGCAAGTTGGGGAAGCCGTATCCGTTGCTGAATCTGGCATATGTGGGAGTAT
AAGACGCGCAGCGTCGCATCAGGCATTTTTTTCTGCGCCAATGCAAAAAGGCCATCCGTCAG
GATGGCCTTTTCGCATAATTTGATGCCTGGCAGTTCCTACTCTCGCATGGGGAGACCCACACA
CTACCATC* NM:i:0 MD:Z:153168 AS:i:153168 XS:i:0

NODE_6_length_153168_cov_18.165897: Read ID, indicating the node from an assembly, its length (153168 base pairs), and coverage (18.17x).

16: Mapping flag, indicating the read is aligned in the reverse direction.

NC_000913.3: The reference sequence (chromosome) where the read aligns.

2572950: The position on the reference genome where the read aligns.

60: Mapping quality, indicating high confidence in this alignment.

153168M: Read length (153168 base pairs) aligned perfectly to the reference (this is a large alignment, meaning this is likely a long contig).

CACTGGCAAGCGCGATATCCGCCAGCTGCGCCGCCGTCGGCTGTGGCACCACGCTGCAATCG
GCAAAGCCCAACGCCGGGCCGCTGTACTGTGGCAACATCAGGAAAATGGAAGAGAGCGTTT
TACAGCCTGGCTGCAAGCCGATAATGCGTAATCCGGCACGCAGCACATTTGCCGTGGAAGA
GAGGTTGCCCCGCGATACAGACATCCGCTTTACCGGCGCTGACCATTGCTGCGGCGAACATCA
GAGGGTCGGTAAGTTTTTCCAGCGCATCCGGCGGCGTTTTTTCGCCCCGCGCGGGCCAGCCAG
CGATGAGCAAATTCTT...GGTTATTGAGGAGCTTAAGGATAAATTTCTGGTAAGGAGGACACG
TATGGAAGTGGGCAAGTTGGGGAAGCCGTATCCGTTGCTGAATCTGGCATATGTGGGAGTAT
AAGACGCGCAGCGTCGCATCAGGCATTTTTTTCTGCGCCAATGCAAAAAGGCCATCCGTCAG
GATGGCCTTTTCGCATAATTTGATGCCTGGCAGTTCCTACTCTCGCATGGGGAGACCCACACA
CTACCATC: Sequence of the contig.

NM:i:0: No mismatches in the alignment (perfect alignment).

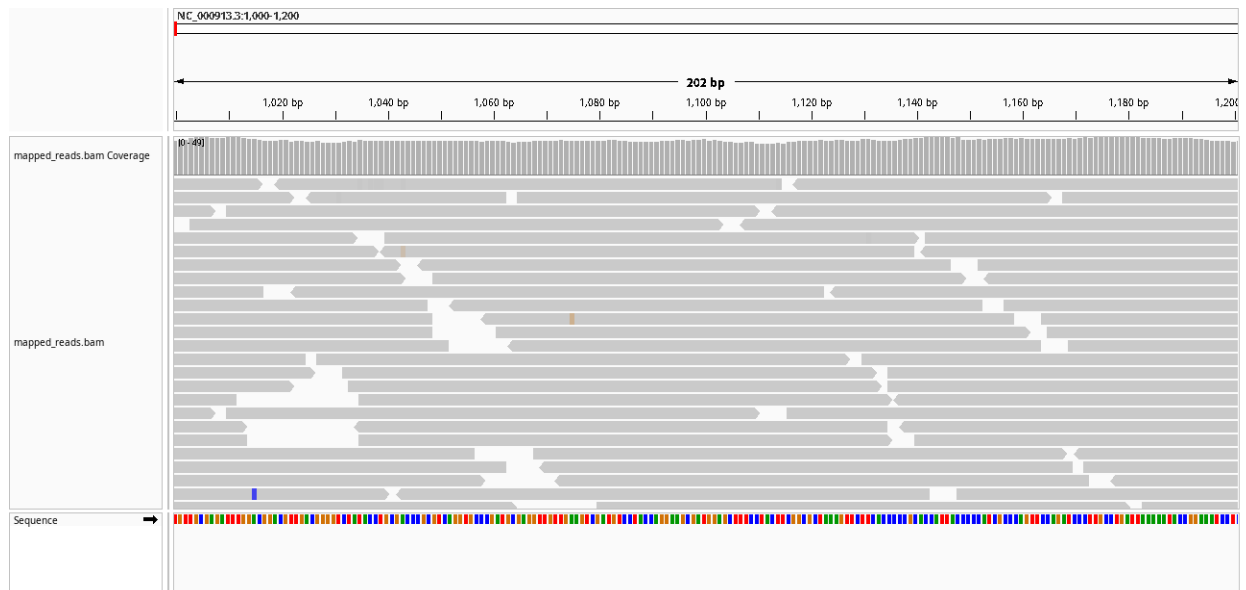
MD:Z:153168: Entire read aligned perfectly (153168 bases aligned with no mismatches).

AS:i:153168: Alignment score (perfect alignment with score 153168).

XS:i:0: No alternative alignment.

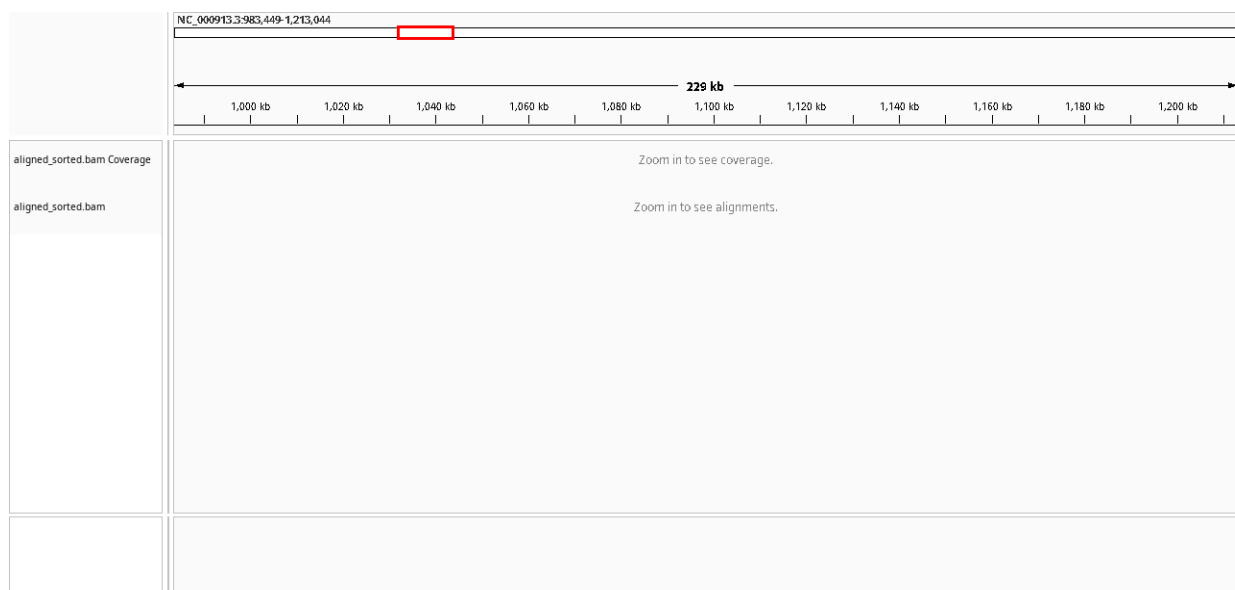
- Use the Integrative Genomics Viewer (IGV) to visualize the mapped reads in a 200-b genomic region of your choice. Select the reference genome fasta and GTF file (GTF is optional).

Mapping raw sequencing reads to a reference genome:



The figure shows the mapping of raw Illumina sequencing reads to the reference genome NC_000913.3 in *E. coli*. The alignments were visualized using the Integrative Genomics Viewer (IGV). The high density of aligned reads in this region suggests good sequencing depth and coverage. This indicates that the reads were successfully aligned to the reference, with minimal mismatches or gaps. The alignment confirms the accuracy of the sequencing process and the quality of the data used for further analysis.

Alignment Results of the Assembled Genome with the Reference Genome:



The figure shows the alignment of the de novo assembled genome to the reference genome NC_000913.3 in *E. coli*. The alignments were visualized using the Integrative Genomics Viewer (IGV). The continuity and coverage of the assembled contigs in this region demonstrate the successful assembly of the genome.

The high concordance between the assembled contigs and the reference genome highlights the accuracy of the assembly process.

- Determine the percentage of short reads that are mapped to the reference genome. Hint: use samtools flagstat.

Mapping raw sequencing reads to a reference genome:

2261059 + 0 in total (QC-passed reads + QC-failed reads)

2260675 + 0 primary

0 + 0 secondary

384 + 0 supplementary

0 + 0 duplicates

0 + 0 primary duplicates

2258059 + 0 mapped (99.87% : N/A)

2257675 + 0 primary mapped (99.87% : N/A)

0 + 0 paired in sequencing

0 + 0 read1

0 + 0 read2

0 + 0 properly paired (N/A : N/A)

0 + 0 with itself and mate mapped

0 + 0 singletons (N/A : N/A)

0 + 0 with mate mapped to a different chr

0 + 0 with mate mapped to a different chr (mapQ>=5)

Alignment Results of the Assembled Genome with the Reference Genome:

266 + 0 in total (QC-passed reads + QC-failed reads)

262 + 0 primary

0 + 0 secondary

4 + 0 supplementary

0 + 0 duplicates

0 + 0 primary duplicates

264 + 0 mapped (99.25% : N/A)

260 + 0 primary mapped (99.24% : N/A)

0 + 0 paired in sequencing

0 + 0 read1

0 + 0 read2

0 + 0 properly paired (N/A : N/A)

0 + 0 with itself and mate mapped

0 + 0 singletons (N/A : N/A)

0 + 0 with mate mapped to a different chr

0 + 0 with mate mapped to a different chr (mapQ>=5)

- Get the read depth for the sorted BAM file at all positions of the reference genome and report the mean of all reads. Hint: use samtools depth.

Mapping raw sequencing reads to a reference genome:

Mean read depth: 48.4198

Alignment Results of the Assembled Genome with the Reference Genome:

Mean read depth: 1.00109

- Make yourself familiar with the CIGAR format. How do you interpret the following expressions?

29S21=1X25=

29S: 29 bases are soft-clipped (not aligned).

21=: 21 matches (aligned to the reference).

1X: 1 mismatch.

25=: 25 matches (aligned to the reference)

20M2I1M1D10M

20M: 20 matches.

2I: 2 insertions.

1M: 1 match.

1D: 1 deletion.

10M: 10 matches

5M10N25M

5M: 5 matches.

10N: 10 bases skipped (spanning a large gap in the reference).

25M: 25 matches